

StaticSense: Static Object Detection using Wi-Fi Channel State Information

Sulav Gaire

Institute of Engineering, Pashchimanchal Campus,
Tribhuvan University
Pokhara, Nepal
sulavgaire@gmail.com

Aayush Shrestha

Department of Computer Science and Engineering,
University of South Carolina,
Columbia, USA
aayushstha54321@gmail.com

Bibek Chand

Institute of Engineering,
Advanced College of Engineering & Management,
Tribhuvan University
Kathmandu, Nepal
chandbibak09@gmail.com

Nabin Lamichhane

Institute of Engineering, Pashchimanchal Campus,
Tribhuvan University
Pokhara, Nepal
babunabin@wrc.edu.np

Abstract—Wireless sensing with Wi-Fi Channel State Information (CSI) has proven effective for recognizing dynamic activities. However, detecting static objects is still difficult because their effects on the wireless channel are subtle and can be masked by environmental clutter. In this work, we present a method for detecting and classifying static objects inside shielded enclosures using low-cost ESP32 microcontrollers in a single-input, single-output (SISO) setup. To reduce interference, we enclose the devices in a Faraday-cage-like box and process the CSI data by subtracting empty-box baselines, sanitizing phase with linear detrending, and applying Hampel filtering to clean amplitude variations. These processed features are then input to a ResNet-18 model. Our experiments, conducted at 100 Hz, achieve 98.27% accuracy across five object types and the empty baseline, with high precision (0.9828) and F1-score (0.9827). The findings demonstrate that CSI-based sensing can reliably detect static objects. In conclusion, this technique of using Wi-Fi-CSI for static object detection offers a practical way to support IoT applications such as contraband detection and security screening, without relying on complex systems.

Index Terms—Wi-Fi sensing, Channel State Information (CSI), static object detection, ESP32, deep learning, ResNet-18

I. INTRODUCTION

Wireless sensing has evolved as a powerful, non-intrusive paradigm for monitoring environments and activities [1], [2]. Leveraging ubiquitous wireless signals, particularly Wi-Fi Channel State Information (CSI), enables a wide range of applications, including human activity recognition (HAR) [3], indoor localization and tracking [4], gesture recognition [4], [5], and health monitoring, such as respiration tracking [4]. CSI, extracted from the Physical (PHY) layer of modern Wi-Fi systems, provides fine-grained information about the signal propagation path, capturing scattering, multipath reflection, diffraction, fading, and power decay across multiple subcarriers [5]. This level of detail allows the capture of subtle changes in the physical environment [3].

Even though detecting dynamic phenomena (e.g. movement or gestures) has obtained great success, most approaches rely

on time-dependent signal traits such as Doppler shifts and short-term CSI amplitude or phase fluctuations [6]. However, a static object shows only a subtle, constant multipath distortion to the CSI signal. Detecting this subtle, fixed disturbance is quite challenging because the static signal component is easily confounded by environmental clutter (e.g., walls and furniture) and device-specific noise, such as carrier frequency offsets (CFO). Indeed, some robust systems designed for dynamic HAR explicitly focus on removing static components to improve generalization across environments. Consequently, methods developed for dynamic detection are often inadequate for the detection of static objects.

This research is motivated by the lack of prior literature that clearly describes a procedure to detect static objects using commodity Wi-Fi Channel State Information effectively and reliably. This capability is often hidden by environmental nuisances and hardware noise. To address the gap, we propose a novel methodology that uses signal processing to extract subtle signal distortions and deep learning techniques to classify them. These methods are carefully designed to isolate the subtle signal distortions caused by static objects from surrounding noise. Our approach focuses specifically on extracting and analyzing subtle, static multipath distortions in the CSI amplitude profiles. We utilize techniques such as baseline subtraction of empty-environment CSI signatures to isolate the pure object-induced effects, thereby enabling high accuracy without requiring object motion. We achieve practical, unobtrusive sensing using only commodity Wi-Fi devices.

The proposed system utilizes low-cost ESP32 microcontrollers in a Single-Input Single-Output (SISO) configuration to capture the CSI data. This choice was made to enable enabling low-power, cost-effective sensing solutions for edge computing and IoT ecosystems. The signal processing pipeline is used to convert raw Wi-Fi CSI data into features that deep learning architectures (such as ResNet-18) are optimised well

for the classification of static objects.

The key contributions are:

- A validated methodology for static object detection and classification with Wi-Fi CSI.
- A novel experimental setup for designing a Faraday cage-like an enclosure (aluminium foil-wrapped metallic box) with low-cost commercial hardware in a simple SISO configuration, enabling high-fidelity isolation from ambient environmental noise for static sensing tasks.
- A rigorous signal processing technique utilising a baseline subtraction pipeline that isolates only the static object-induced CSI distortions.
- Achieved 98 percent accuracy across multiple static metallic object classes, demonstrating the feasibility of a ResNet-18 model.

II. RELATED WORK

Static object detection using Wi-Fi Channel State Information poses a unique challenge that really sets it apart from the more commonly researched area of dynamic human sensing [6]. While the movement of human subjects allows dynamic systems to rely on time-varying effects like the *micro-Doppler shift* [6], a static object only introduces a *constant offset* in the CSI signal, making it difficult to differentiate from static clutter or ambient environmental changes [6]. Prior work highlights an important limitation. Doppler's traces, which proves effective for human activity recognition (HAR), are largely insensitive while detecting static objects. As a result, they fail to reliably detect the presence of an immobile person [6].

A. Dynamic Sensing and General Applications of CSI

Most of the CSI-based research focuses on dynamic studies and relies on commodity Wi-Fi hardware.

- **Human Activity Recognition (HAR):** Numerous systems employ CSI amplitude and phase information to train deep learning models such as 2D-CNNs, LSTMs, and BLSTMs for classification [3], [7]. For instance, SHARP [6] achieves environment & person independent HAR by utilising a phase sanitisation technique that extracts micro-Doppler features from CSI data. This approach displays high accuracy in distinguishing dynamic activities, for example, walking and running. Also, systems like DeepSense [8] and CARM [9] utilize CSI for HAR.
- **Localization and Tracking:** Pilot [10] and similar systems capitalise on the temporal stability and frequency diversity of CSI to create effective passive localization indoors. This technique outperforms less sophisticated RSS-based methods exemplified by Nuzzer [10]. Similarly, CrossSense system addresses cross-site generalization for gait identification and gesture recognition tasks by leveraging machine learning to synthesize the necessary training samples needed when deploying the system in a new physical environment. [1].

- **Fine-Grained Detection:** WiDraw is a fine-grained system that recognises gestures by using the angle of arrival model [11]. SiWiS is a fine-grained system that utilises a new RF sensing hardware add-on to achieve phase-coherent sensing like Doppler radars for estimating human pose and segmenting body masks [12].

B. Static Object and State Detection

Although static object detection using Wi-Fi is challenging, some systems have addressed static states or states of everyday objects [4]:

- **Indoor Object State Changes:** Ohara et al. proposed detecting state changes (e.g., open/close events of doors and windows) of indoor objects using commodity Wi-Fi [13]. This system employs a processing pipeline that includes Independent Component Analysis (ICA) for blind source separation, Convolutional Recurrent Neural Networks (RNNs) for feature learning, and Hidden Markov Models (HMMs) based on object state transition knowledge. ICA was specifically tailored to extract segments related to the events of interest to maximize independence [13].
- **Novel RF Imaging:** Object imaging using RF signals has been explored in the past. That usually demands complicated methods like Synthetic Aperture Radar (SAR) or high-frequency systems like mm-wave radars (suggested for static pose recognition) [6]. Recent research has proposed the use of Reconfigurable Intelligent Surfaces (RIS) in conjunction with Wi-Fi Access Points to assist in imaging systems [14].

C. Proposed Approach and Enabling Technology

Our research aligns with the recent trend toward low-cost, edge computing solutions. We achieve this via two key aspects:

- **Low-Cost Hardware:** We use simple, inexpensive ESP32 microcontrollers in a basic Single Input Single Output (SISO) configuration, enabling standalone sensing right at the edge [4]. However, prior systems depended on powerful devices that utilize specialised Software-Defined Radios (SDR) or Intel 5300 NICs [7].
- **Environment Isolation and Processing:** To suppress environmental noise that usually masks subtle static effects, our approach employs a novel setup using a Faraday cage-like enclosure and with rigorous baseline subtraction. This process isolates minute static multipath distortions created by static objects in the CSI amplitude profiles, which are then effectively classified using a ResNet-18 deep learning architecture well-suited for interpreting this complex data.

III. METHODOLOGY

This section presents a detailed methodology applied in detecting static objects using Wi-Fi channel state information in a controlled environment using Low-cost hardware (ESP32) and a tailored signal processing technique.

A. System Overview and Architecture

The core idea is detecting multipath alterations in the CSI profile in an entirely non-invasive, passive sensing approach caused by a static object. This is different from dynamic sensing, as they depend on Doppler shifts caused by movement. We overcome the prior challenge of environmental clutter by integrating a unique hardware setup and advanced signal processing.

1) *Hardware Setup*: Two ESP32-WROOM-32 modules are configured in SISO mode. The Tx broadcasts packets via ESP-NOW at 100 Hz on channel 11 (2.4 GHz). The Rx captures CSI using the ESP-IDF API. Both are placed inside a $20 \times 20 \times 6$ cm cardboard box wrapped in aluminum foil (Faraday cage). Tx/Rx antennas are fixed 6 cm apart, with objects placed inside the box in front of them. Faraday cage-like enclosure, is essential for achieving high-fidelity isolation from ambient environmental noise to successfully capture the subtle, static CSI distortions. This environment is a controlled condition that was necessary to validate the signal processing methodology

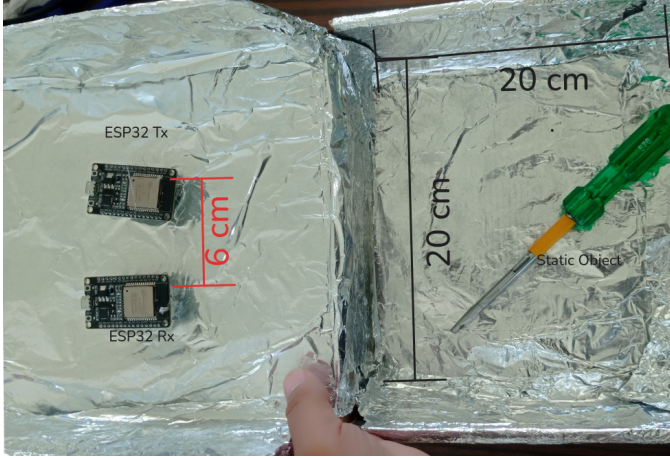


Fig. 1: Hardware setup.

2) *Software Setup*: The Tx firmware transmits packets at MCS0 with Short Guard Interval. The Rx firmware logs 384-byte CSI buffers to a host PC via USB serial. A Python Flask server stores data in structured format (CSV and MAT).

B. Data Collection and Ground Truth

CSI was collected in two phases: (i) **baseline sessions**: CSI was recorded with the enclosure empty to establish the environmental reference., and (ii) **object sessions** CSI was recorded with a single static object (motor, pliers, scissor, screwdriver, or tweezers) placed inside the enclosure. For each class, 10–20 sessions of 20s were recorded. Objects were varied in position/orientation to capture multipath diversity.

C. Signal Characteristics and Processing

In OFDM, CSI per subcarrier k channel response is given as:

$$H(f_k) = |H(f_k)|e^{j\phi(f_k)} \quad (1)$$

Static objects in the environment affect both components of the channel:

- **Amplitude**: Altered via absorption and scattering.
- **Phase**: Changed due to variations in path length, given by:

$$\Delta\phi = \frac{2\pi\Delta d}{\lambda} \quad (2)$$

where Δd is the change in path length and λ is the signal wavelength.

Processing Pipeline The collected CSI packets undergo the following processing steps:

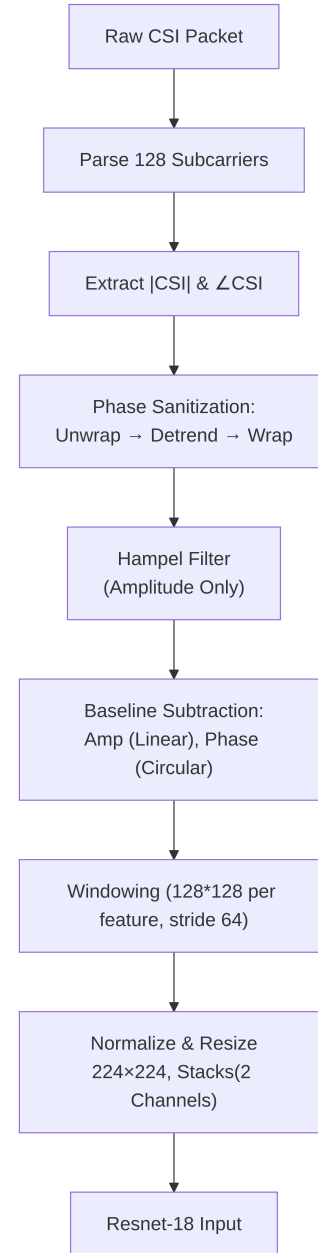


Fig. 2: Signal processing pipeline.

- 1) Parsing: Extract 128 complex values from HT-LTF fields (subcarriers ≈ -64 to 63).
- 2) Feature Extraction: Compute amplitude $|H|$ and phase $\phi = \angle H$.
- 3) Phase Sanitization (per packet):
 - a) Unwrap phase.
 - b) Perform linear regression against subcarrier indices k to remove Carrier Frequency Offset (CFO) & Sampling Frequency Offset (SFO) errors as phase can be messy because of hardware errors or timing mismatches (CFO and SFO)::

$$\phi_{\text{unwrapped}} = \phi_{\text{CFO}} + 2\pi \delta t k + \phi_{\text{true}} \quad (3)$$

where δt is the timing offset.

- c) Subtract the fitted line:

$$\phi_{\text{sanitized}} = \phi_{\text{unwrapped}} - (\text{slope} \cdot k + \text{intercept}) \quad (4)$$

- d) Wrap back into $[-\pi, \pi]$:

$$\phi_{\text{wrapped}} = \angle e^{j\phi} \quad (5)$$

Linear detrending reduces SFO-induced errors reducing phase noise, and wrapping preserves the angular nature of phase without discontinuities.

- 4) Outlier Removal: Apply Hampel filter (window $k = 7$, threshold $t_0 = 3$) to amplitudes across time per sub-carrier. Phases are left unfiltered to avoid distortion delay information; sanitization suffices. This handles Wi-Fi burst noise effectively while preserving delay information.
- 5) Baseline Subtraction: Compare the current object scene with a reference empty scene.
 - Amplitude residual:

$$A_{\text{residual}} = \max(A_{\text{object}} - \bar{A}_{\text{empty}}, 0) \quad (6)$$

- Phase residual using circular mean:

$$\bar{\phi}_{\text{empty}} = \angle \left(\frac{1}{N} \sum e^{j\phi_{\text{empty}}} \right) \quad (7)$$

$$\phi_{\text{residual}} = \angle \left(e^{j(\phi_{\text{object}} - \bar{\phi}_{\text{empty}})} \right) \quad (8)$$

Circular statistics handle phase periodicity and capture relative shifts due to objects; vector averaging ensures robust differencing. This gives the change caused by objects, ignoring background.

- 6) Windowing and Normalization: Segment the data into windows (size = 128, stride = 64). Amplitudes are normalized to $[0, 1]$, and phases are mapped from $[-\pi, \pi]$ to $[0, 1]$. The resulting matrices are resized to 224×224 and stacked as a two-channel input “spectrogram” (magnitude + phase) for CNNs.

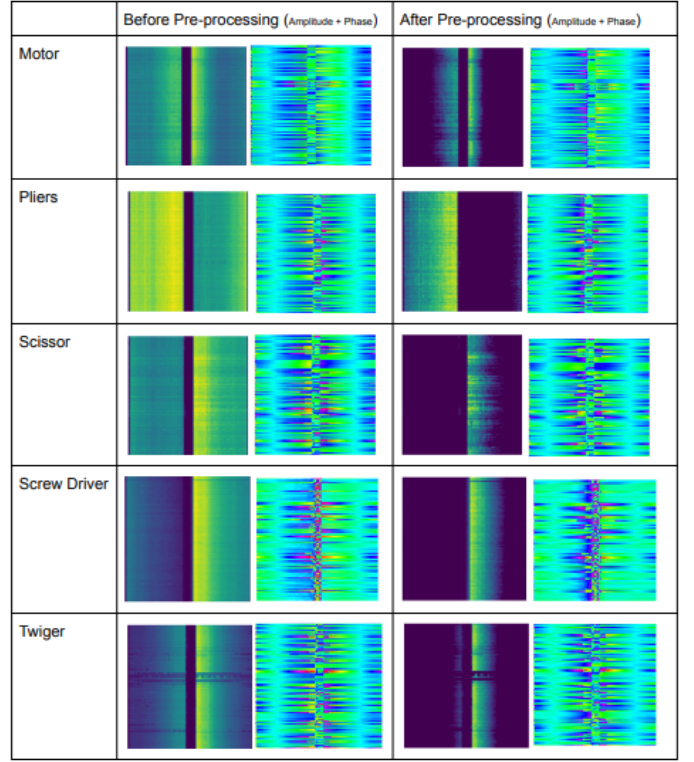


Fig. 3: Before and after signal processing

D. Classification and Model Training

Preprocessed CSI was converted into two-channel images (amplitude, phase). A ResNet-18 CNN was adapted for two-channel input and five-class output. Training used Adam optimizer ($\text{lr}=1 * e^{-4}$, weight decay= $1 * e^{-4}$), cross-entropy loss function, batch size=16, for 50 epochs. Dataset was split 70/30 with weighted sampling and augmentation (horizontal flips, small rotations).

IV. RESULTS

The proposed model demonstrates strong and consistent performance across all object categories. Table I summarizes the overall evaluation metrics obtained on the test dataset.

TABLE I: Performance Evaluation Metrics

Metric	Value
Accuracy	0.9827
Precision	0.9828
Recall	0.9827
F1 Score	0.9827

A. Training and Validation Performance

Figure 4 shows the training and validation loss and accuracy over 50 epochs. The training loss decreases steadily. Meanwhile, the validation loss shows a similar pattern with some ups and downs. This suggests it generalises well. Both accuracy curves level off above 95%. This indicates that the network learns important features without overfitting.

V. CONCLUSION

Finally, we present a novel static object detection system that leverages Wi-Fi channel-state information in a controlled lab setting. By integrating baseline subtraction, phase sanitisation, and a deep-learning technique, over 98% accuracy was achieved for classifying across 5 object classes. This approach may prove promising for security checks and inventory tracking.

However, this research acknowledges two main constraints. First, to mask the environmental dynamic noise and isolate the subtle signal change induced by the static object, the system needs an EMI shielded environment, in our case, the Faraday cage. While this was a necessity to validate that we can effectively detect static objects using channel state information, real-world deployments still remain the main challenge, where dynamic ambient noise and Doppler shifts are predominant. Further research could focus on effective signal processing and deep learning techniques that could suppress predominant dynamic noise and isolate and classify subtle signal changes induced by the static object. This may mean leveraging directional antennas that direct signals right at the object, or it can also be advanced spatio-temporal filtering.

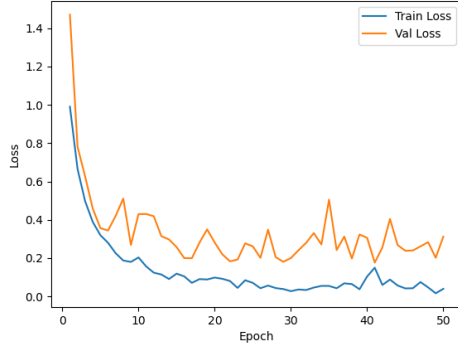
Second, this was not deliberate, but we noticed that most objects we picked from our lab were mostly metal or had metal parts. We could classify them well; this proves that this process can sense material that interacts with the Wi-Fi channel state information. Future studies are needed to determine the technique's efficacy against purely non-metallic objects.

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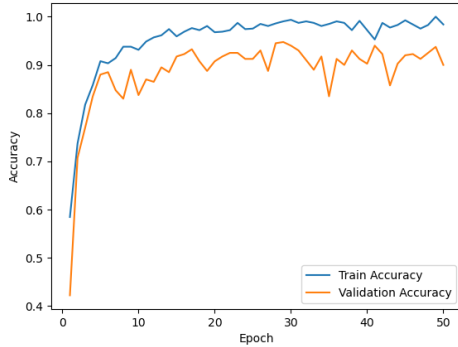
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(a) Training and validation loss.



(b) Training and validation accuracy.

Fig. 4: Model performance over 50 epochs.

B. Confusion Matrix Analysis

The confusion matrix in Fig. 5 illustrates the classification results for each class. The model achieves almost perfect separation between classes, with just a few errors in determining *pliers* and *scissors*. This may be due to the visual similarity of these tools. Overall, the results confirm that the network consistently tells different objects apart with high accuracy and recall.

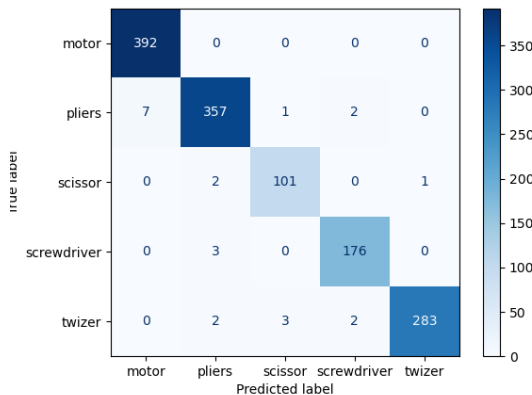


Fig. 5: Confusion matrix for object classification.

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